

9XAI HACKATHON INSTRUCTION SET

Building the Digital Twin of the Minister of Public Sector Development

1. Purpose of This Hackathon

This is a 3.5 day, 28-hour hackathon built around seven four-hour sprints.

Your mission is to design and progressively build the future digital twin of the Minister of Public Sector Development through seven core capabilities.

You are not building isolated demos.

You are building a serious AI-native ministerial intelligence system made of modular capabilities that can work independently and together.

The strongest outcomes may later be combined into one broader ministerial digital twin architecture.

This hackathon is also designed to accelerate how you operate as a team. It is a test of execution, ownership, adaptability, and peak collaboration under pressure.

2. What Is the Digital Twin in This Challenge

For this challenge, the digital twin means:

- A bilingual Arabic and English intelligence layer that helps the Minister and stakeholders:
- Understand what is happening
- See risks and opportunities early
- Improve services
- Coordinate across institutions
- Assess readiness for reform
- Evaluate policy implications
- Hear the voice of citizens clearly
- Operate with speed and confidence

- Move from insight to action
- It must communicate professionally in Arabic and English.

3. Core Challenge

Build the seven mandatory capabilities of the Minister's digital twin.

Each capability must include:

- Its own agentic team
- Its own org structure
- Clear workflows
- Guardrails
- Pipeline logic
- Microservices thinking
- Ability to integrate with other capabilities
- You may also build additional capabilities beyond the seven required ones.

4. Mandatory Technology Principles

Use:

- Paperclip as the orchestration layer
- Gemma 4 as the reasoning and execution model

Your design should reflect:

- Agentic teams
- Modular architecture
- Pipeline workflows
- Microservices mindset
- Guardrails
- Reusable components
- Scalable logic

5. Team Format

- This is a pair challenge.
- Each pair works independently.

- Both participants are expected to contribute across discovery, design, building, testing, iteration, and presentation.
- This is not a divide-the-work exercise.
- The highest-performing pairs will behave like a small elite unit with shared ownership, fast trust, high standards, and continuous support for one another.

6. The Seven Mandatory Capabilities

1. Executive Radar

What this capability is

This is the Minister's high-level line of sight. It helps her understand, in one place, what is moving well, what is slipping, what is becoming risky, and where intervention is needed now.

Why this capability matters

A minister leading modernization cannot rely on scattered updates, late reporting, or fragmented institutional narratives. She needs an intelligent layer that turns noise into attention. The modernization agenda itself spans multiple pillars and entities, which makes early visibility essential.

What it should help the Minister do

- See the few issues that matter most right now.
- Spot slippage early.
- Recognize patterns before they become problems.
- Separate signal from noise.
- Decide where to intervene, escalate, or stay out.

Examples of what it may monitor

- Modernization initiatives drifting off timeline.
- Entities that look green on paper but weak in reality.
- Emerging delivery bottlenecks.
- Execution risks that are spreading across more than one institution.
- Windows of opportunity where ministerial sponsorship can unlock momentum.

Strong outputs it can produce

- A ministerial morning brief.
- A top issues watchlist.
- An early warning alert.
- A momentum and slippage snapshot.

- A “needs intervention now” view.

2. Service Friction Intelligence

What this capability is

This capability helps the Minister understand where public services are still painful, confusing, slow, repetitive, or unnecessarily bureaucratic for citizens, businesses, and front-line staff.

Why this capability matters

Public sector modernization becomes real when service experience improves. Jordan’s roadmap puts strong weight on citizen-centered service quality, simplification, governance of service development, and responsiveness to changing needs and expectations.

What it should help the Minister do

- See where the public loses time, trust, or energy.
- Detect the services that create the most friction.
- Understand whether the problem is policy, process, handoff, system design, or poor service governance.
- Prioritize which services deserve redesign first.

Examples of what it may uncover

- A service with too many approval steps.
- A process that forces the same document to be submitted multiple times.
- A digital service that exists but still pushes the user back into manual work.
- A high-volume public service where waiting time is not the real problem, but unclear instructions are.
- A citizen journey that crosses several entities and breaks at the handoff.

Strong outputs it can produce

- A service pain map.
- A friction diagnosis by service journey.
- A shortlist of services that should be redesigned first.
- A quick-win simplification plan.
- A “root cause of friction” summary for leadership.

3. Institutional Readiness Analyzer

What this capability is

This capability helps the Minister assess whether a ministry, department, or public entity is actually ready to execute modernization well, not just claim alignment with it.

Why this capability matters

Reform does not fail only because the strategy is weak. It often fails because institutions are not equally ready in governance, operating model, talent, leadership, decision speed, or execution discipline. The modernization agenda itself includes organizational structure, governance, human resources, and institutional effectiveness as core themes.

What it should help the Minister do

- See which entities are ready to move fast.
- Identify where capability gaps are likely to derail reform.
- Distinguish between institutions that need empowerment and institutions that need deeper restructuring.
- Sequence reform more intelligently.

Examples of what it may assess

- Whether the entity has clear ownership for modernization.
- Whether decision-making is too slow or too centralized.
- Whether required capabilities exist in the team.
- Whether the organizational model supports reform or quietly blocks it.
- Whether the institution can absorb new systems, processes, and accountability demands.

Strong outputs it can produce

- An institutional readiness scorecard.
- A maturity heatmap.
- A capability gap diagnosis.
- A readiness-by-entity comparison.
- A targeted readiness improvement plan.

4. Policy Impact Assistant

What this capability is

This capability helps the Minister think through the likely consequences of policy, regulatory, or reform choices before implementation and then track what those choices are actually producing afterward.

Why this capability matters

The modernization agenda is not only about services and structures. It is also about legislation, policy quality, impact assessment, compliance, accountability, and more evidence-based decision-making. Jordan's public sector modernization work explicitly includes improving policy-making and decision-making through impact assessment and proactive legislative review.

What it should help the Minister do

- Compare options before committing.
- Understand who gains, who carries burden, and where friction may appear.
- Anticipate implementation complexity.
- Challenge decisions that look attractive in principle but weak in execution.

Examples of what it may analyze

- Whether a proposed reform will simplify or complicate service delivery.
- Whether a legislative change creates new compliance overhead.
- Whether one policy option is faster but less equitable.
- Whether an intervention creates burden for front-line units that cannot absorb it.
- Whether an intended reform collides with existing rules, incentives, or institutional realities.

Strong outputs it can produce

- A policy options comparison.
- A likely impact summary.
- A tradeoff brief.
- An implementation risk note.
- A recommended course of action with reasoning.

5. Cross-Entity Coordination Engine

What this capability is

This capability helps the Minister see and manage how work moves across ministries, agencies, units, and public bodies when no single entity can deliver the result alone.

Why this capability matters

A large share of public sector failure does not come from bad intent. It comes from dependency blindness, unclear ownership, weak handoffs, unresolved blockers, and the inability of institutions to operate as one unit. Jordan's modernization work explicitly emphasizes better governance, implementation structures, central monitoring, and coordinated execution.

What it should help the Minister do

- See where coordination is breaking.
- Understand who depends on whom.
- Spot unresolved blockers early.
- Clarify ownership when work spans multiple actors.
- Strengthen execution across the system, not just inside one entity.

Examples of what it may manage

- A service reform that depends on three ministries moving in sequence.
- A modernization initiative stalled because one entity has not delivered its part.
- Repeated duplication because two entities are solving overlapping problems separately.
- Escalations that should have happened earlier but did not.
- Shared national priorities where progress in one institution is blocked by another.

Strong outputs it can produce

- A dependency map.
- A blocker and escalation dashboard.
- A cross-entity action tracker.
- A shared ownership matrix.
- A coordination health summary for leadership.

6. Citizen Voice Translator

What this capability is

This capability turns the voice of the public into usable leadership intelligence. It does not merely collect complaints. It translates patterns in citizen and stakeholder feedback into structured insight that can guide action.

Why this capability matters

Citizens often see service failures, policy confusion, procedural pain, and trust erosion before these issues appear in formal reporting. A serious modernization agenda has to hear those signals early and convert them into something decision-makers can use. Jordan's roadmap is explicitly citizen- and stakeholder-centered.

What it should help the Minister do

- Understand what people are really struggling with.
- See recurring themes rather than isolated anecdotes.
- Distinguish emotional noise from meaningful patterns.
- Connect public voice to service, policy, or institutional action.

Examples of what it may process

- Citizen complaints around one recurring service problem.
- Feedback showing that a procedure is technically available but practically unusable.
- Growing frustration around unclear eligibility, requirements, or delays.
- Patterns in stakeholder concerns from the private sector or civil society.
- Signals that trust is improving or deteriorating in a specific area.

Strong outputs it can produce

- A public voice brief.
- A recurring issues summary.
- A citizen trust pulse.
- A “what the public is telling us” synthesis.
- Action recommendations rooted in citizen signal patterns.

7. Ministerial Chief of Staff Office

What this capability is

This is the Minister’s execution brain and operating companion. It is the capability that helps translate priorities, meetings, decisions, and follow-up into disciplined movement.

Why this capability matters

Insight without execution is decoration. A minister needs a system that helps maintain focus, prepare well, convert discussions into action, track commitments, and preserve continuity from one day and one week to the next.

What it should help the Minister do

- Start each day with clarity.
- Enter meetings briefed and prepared.
- Turn decisions into owned next steps.
- Track unresolved commitments.
- Maintain a coherent rhythm across a demanding agenda.

Examples of what it may support

- Preparing the Minister before a key modernization meeting.
- Summarizing the decisions made and what must happen next.
- Tracking follow-up across multiple stakeholders.
- Reprioritizing the Minister’s agenda when new issues emerge.
- Preserving continuity so important threads do not disappear between meetings.

Strong outputs it can produce

1. A ministerial daily brief.
2. A meeting prep pack.
3. A decision memo.
4. A follow-up tracker.
5. A weekly priorities and commitments view.

One important instruction to fellows

These seven capabilities are not seven disconnected tools.

Each one should be built as a distinct agentic capability with its own team, logic, workflow, and outputs. At the same time, each capability should be designed so it can eventually plug into a larger bilingual ministerial digital twin that works in both Arabic and English, serves multiple stakeholders professionally, and operates as one integrated intelligence system. The seven capabilities themselves are grounded in Jordan's modernization agenda across services, institutions, legislation, governance, and citizen-centered reform.

7. Why This Hackathon Will Leapfrog Your Team Skills

This challenge is intentionally designed to accelerate how you work together.

Ownership

- You cannot hide inside a long process or wait for instructions.
- Pairs that win will take initiative, solve problems early, and own outcomes from start to finish.

Peak Performance Under Pressure

- Short sprints create urgency. Urgency reveals habits.
- Strong teams learn to stay calm, prioritize clearly, and keep moving.

Dream Team Dynamics

- High-performing pairs usually show:
- Fast trust
- Clear communication
- Mutual respect
- Complementary strengths
- Rapid conflict resolution
- Shared standards
- Energy under pressure

Real Collaboration

- This is not one person building while the other watches.
- The best teams think together, challenge each other, improve each other, and raise the final output together.

Learning Velocity

- Because you must build while learning, your growth speed increases.
- You will likely learn more in this format than in slow theoretical settings.

8. Sprint Model

Total Duration

3.5 days

7 sprints

4 hours each

28 total working hours

Important Rule

- You choose the order in which you build the seven capabilities.
- This is part of the challenge.
- Your sequence should be intentional, not random.
- You should be able to explain:
 - Why you chose that order
 - What dependencies shaped your decision
 - What gave you the highest leverage first
 - What tradeoffs you made
- Different teams may choose different paths.
- That is expected.

9. What Happens in Every Sprint

Every sprint should include lightweight versions of:

- Discover
- Decide

- Build
- Test
- Learn
- Improve

Every sprint must end with visible progress.

No sprint ends empty.

10. Sprint Checkpoints

At the end of each sprint, every pair must be ready to present a short pulse update.

You Must Show

1. Capability Chosen This Sprint

Which capability did you work on and why now

2. Decision Logic

Why this capability came first, now, or next in your sequence

3. What Was Built

Prototype progress, workflows, outputs, architecture, or integration progress

4. What Improved

What became smarter, clearer, faster, or more useful

5. Challenges Faced

- Technical blockers
- Architecture issues
- Reasoning quality issues
- Time pressure
- Scope confusion
- Integration friction

6. Team Dynamics

- How you worked as a pair
- How you divided and shared ownership
- How you solved disagreements
- How you improved speed or quality together

7. Shipping Status

Did you ship a usable version of this capability

8. What Comes Next

What capability or improvement you plan next and why

11. Pulse Check Timing

Pulse checks happen:

- First thing in the morning
- Midday
- End of the day

They are for direction, challenge, momentum, and support.

12. What Must Exist Across All Capabilities

Every capability should reflect:

- Its own agentic org structure
- Clear ownership
- Inputs and outputs
- Pipeline logic
- Guardrails
- Arabic and English outputs
- Integration readiness

13. Final Submission

Your final submission must include:

- All seven mandatory capabilities completed at working prototype level or clearly demonstrated level
- Any bonus capabilities beyond the seven
- Unified architecture view
- Agentic org structures
- Demo of major workflows
- Arabic and English outputs
- Executive summary
- Recommended next phase

14. Evaluation Criteria

You will be evaluated on:

- Quality of all seven capabilities
- Strength of sequencing decisions
- Ability to ship capabilities consistently
- Architecture quality
- Orchestration design
- Usefulness of outputs
- Arabic and English professionalism
- Transferability and scalability
- Integration across modules
- Execution quality
- Team ownership and collaboration
- Originality beyond the seven

15. Awards May Include

- Best Overall Digital Twin
- Best Capability Execution
- Best Strategic Sequencing
- Best Architecture
- Best Bilingual Experience
- Best Orchestration Logic
- Most Useful Public Sector Innovation
- Best Dream Team Performance
- Boldest Expansion Beyond the Seven

16. Final Standard

Do not only build capabilities.

Show that you know what to build first, why it matters, how to work like an elite team, and how to keep shipping under pressure.